CYBERSECURITY LAB PROJECT

Home SOC Lab

Network Topology and Configuration

Oracle VirtualBox | NAT | Internal Network | Host-Only

PREPARED BY

Huzaifa Saeed

ENVIRONMENT

4 virtual machines | 3 virtual networks | Windows host

1. Project Overview

This project involved designing and configuring a virtual network topology for a home cybersecurity lab using Oracle VirtualBox. The lab connects multiple operating system environments through isolated virtual networks, allowing them to communicate while maintaining separate network roles.

The design uses three VirtualBox networking modes: NAT, Internal Network, and Host-Only. Each serves a distinct purpose, providing internet connectivity, private virtual-machine communication, and communication between the host computer and virtual machines.

2. Objectives

- Design a virtual network for a multi-machine cybersecurity lab.
- Configure NAT, Internal Network, and Host-Only adapters.
- Assign consistent IPv4 addresses to lab machines.
- Establish communication between virtual machines on the Internal Network.
- Configure persistent Internal Network addresses on Kali Linux and Parrot Security.
- Validate network connectivity and document the final configuration.

3. Lab Environment

The environment consists of four virtual machines hosted in Oracle VirtualBox:

Virtual Machine	Operating System	Primary Role
Parrot Security	Parrot Security OS	Security testing and analysis
Kali Linux	Kali Linux	Penetration testing and security tools
Ubuntu Server	Ubuntu Server	Linux server administration and services
DC01	Windows Server 2022	Windows server administration and future Active Directory lab

The host computer runs Windows and manages the virtual machines through VirtualBox.

4. Network Architecture

Each virtual machine is configured with three virtual network adapters.

4.1 NAT Network Access

The first adapter uses VirtualBox NAT to provide outbound internet connectivity to the virtual machine. This network supports activities such as installing packages, updating operating systems, and retrieving software. VirtualBox manages the NAT gateway and address assignment.

4.2 Internal Network — SOC-INTERNAL

The second adapter connects each virtual machine to an isolated VirtualBox Internal Network named `SOC-INTERNAL`. The Internal Network uses the `10.10.20.0/24` subnet. It provides a private communication segment for the lab machines.

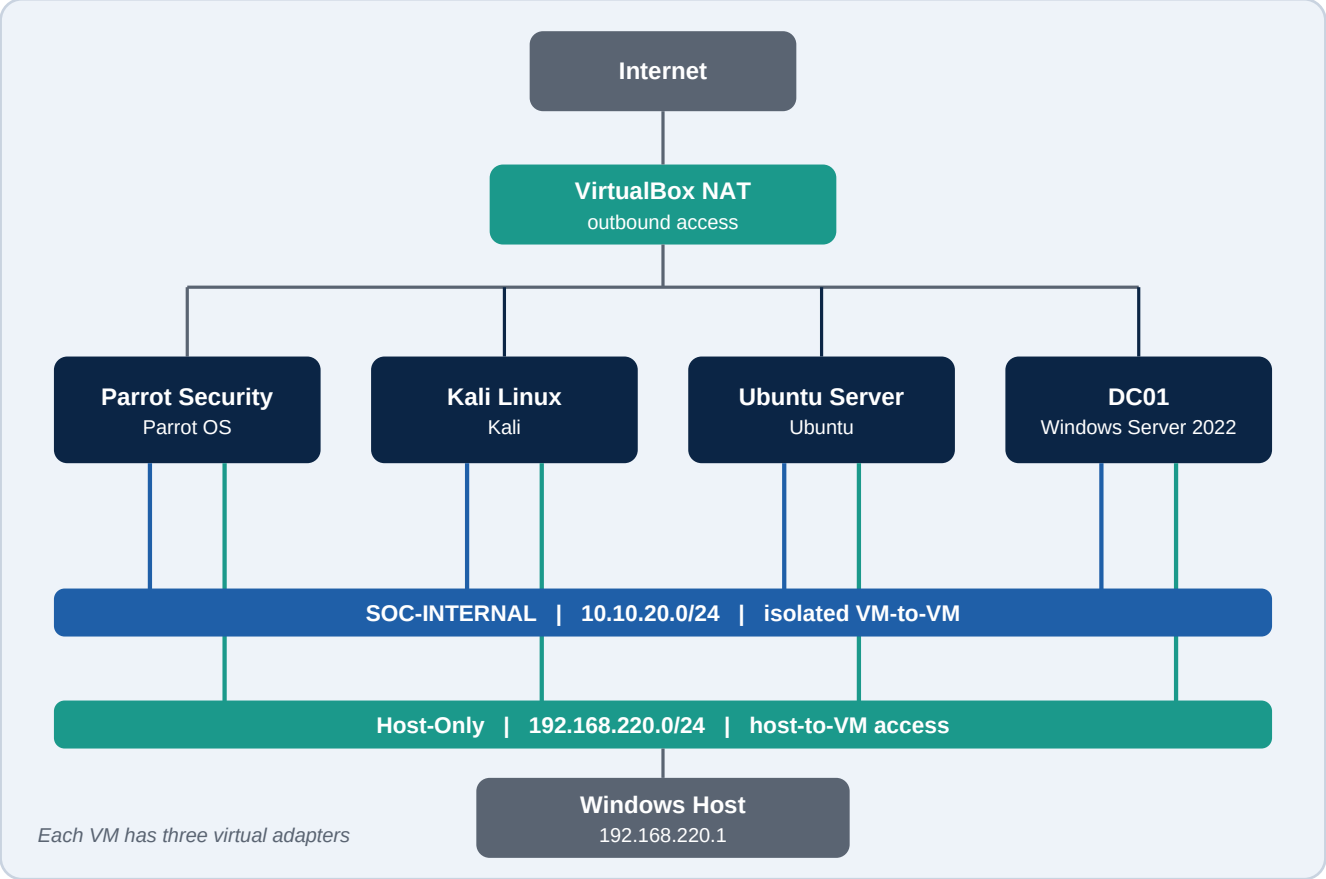
No default gateway is configured on the Internal Network interfaces. Internet access is provided separately through NAT.

4.3 Host-Only Network

The third adapter connects each virtual machine to the VirtualBox Host-Only network. The Host-Only subnet is `192.168.220.0/24`. It allows communication between the Windows host and participating virtual machines without requiring that traffic to pass through the physical network.

The Windows host's Host-Only interface uses `192.168.220.1`.

5. Network Topology Diagram



The diagram represents logical network connections, not physical cabling. Each VM has its own virtual adapters attached to the corresponding VirtualBox networks.

6. IP Address Allocation

The following table records the IPv4 addresses used for the lab configuration.

System	NAT Adapter	Internal Network	Host-Only Network
Parrot Security	`10.0.2.15`	`10.10.20.10/24`	`192.168.220.9/24`
Kali Linux	`10.0.2.15`	`10.10.20.20/24`	`192.168.220.5/24`
Ubuntu Server	`10.0.2.15`	`10.10.20.40/24`	`192.168.220.30/24`
Windows Server 2022 (DC01)	`10.0.2.15`	`10.10.20.50/24`	`192.168.220.40/24`
Windows Host	–	–	`192.168.220.1/24`

The NAT addresses shown are the addresses observed during configuration. VirtualBox NAT commonly assigns the same private address to individual VMs because each VM has its own NAT environment.

7. Persistent IP Configuration

Parrot Security

The Internal Network adapter, `enp0s8`, was configured with the static address `10.10.20.10/24` using NetworkManager. The saved connection profile uses manual IPv4 configuration and does not define a default gateway. The configuration was verified after reboot, and the address remained assigned.

Kali Linux

The Internal Network adapter, `eth1`, was configured with the static address `10.10.20.20/24` using a dedicated NetworkManager connection named `SOC-INTERNAL`. The saved profile uses manual IPv4 configuration and does not define a default gateway. The active connection and assigned address were verified. A separate post-reboot persistence test was not recorded.

Ubuntu Server

Ubuntu Server uses Netplan to configure its network interfaces. Its Internal Network address is `10.10.20.40/24`, and its Host-Only address is `192.168.220.30/24`. The NAT interface obtains its address automatically.

Windows Server 2022

The Windows Server VM, named `DC01`, uses `10.10.20.50/24` on its Internal Network adapter and `192.168.220.40/24` on its Host-Only adapter. Its NAT adapter obtains its address automatically.

8. Connectivity Validation

Connectivity was tested between the Windows Server VM and the other lab machines over the Internal Network. The following communication tests completed successfully:

Source	Destination	Result
Windows Server (DC01)	Parrot Security — 10.10.20.10	Successful
Windows Server (DC01)	Kali Linux — 10.10.20.20	Successful
Windows Server (DC01)	Ubuntu Server — 10.10.20.40	Successful

Successful ping responses confirmed that the tested virtual machines could communicate over the Internal Network at the time of testing.

9. Network Design Considerations

The three-network design separates different types of traffic:

- **NAT:** Provides outbound internet access.
- **SOC-INTERNAL:** Provides private VM-to-VM communication.
- **Host-Only:** Provides a dedicated path between the Windows host and the virtual machines.

The Internal Network operates as a Layer 2 virtual network. VirtualBox provides the virtual network connectivity between attached adapters, so a separate physical switch is not required. A router or default gateway is also unnecessary for communication between devices on the same subnet.

Communication between different subnets would require routing if it becomes part of a future lab design.

10. Resource Management

The host computer has 8 GB of RAM, which limits the practical number of virtual machines that can run simultaneously. Virtual machines were therefore managed individually during configuration and validation to reduce resource pressure. This approach allowed network settings to be configured and tested without unnecessarily running every lab machine at once.

11. Conclusion

The Home SOC Lab network topology was designed and configured using VirtualBox's NAT, Internal Network, and Host-Only networking modes.

The lab uses a dedicated Internal Network subnet for private communication between the virtual machines and a Host-Only subnet for host-to-VM access. Persistent Internal Network IP configuration was established and verified after reboot on Parrot Security, while Kali's saved configuration and active address were verified.

Successful ping tests from Windows Server to Parrot, Kali, and Ubuntu confirmed Internal Network communication during validation.

This topology provides a foundation for future cybersecurity exercises, server administration, network monitoring, and Active Directory lab development.